Experiment 2: Design Zero-shot, One-shot, and Few-shot Prompts to Generate a 200-word Blog on "Applications of Artificial Intelligence in Healthcare"

Aim

To implement Zero-shot, One-shot, and Few-shot prompting techniques using the OpenAI API to generate a 200-word blog on the applications of Artificial Intelligence in Healthcare.

Python Code:

```
from openai import OpenAI

# Paste your OpenAI API Key here
client = OpenAI(
    api_key="sk-proj-
fXs9TwmdiZQU9td92kDIe9wACeeq8S0wsiGwQRfrbEoJq4VGPLU1Np74KCiIODlH8_C
oBWZp8nT3BlbkFJBjvFWrf8sOhvLZ7zO3YVr2MDh2NeIM42jru5-
z2nctrTGHuEgdMAYbFx2lo34b4CY-q8ffMBkA"
)

# ---------- Zero-shot Prompt ----------
zero_shot = """
Write a 200-word blog on 'Applications of Artificial Intelligence in Healthcare'.
"""

response = client.responses.create(
    model="gpt-5.4-mini",
    input=zero_shot
)

print("===== ZERO-SHOT OUTPUT =====")
print(response.output_text)


# ---------- One-shot Prompt ----------
one_shot = """
Example:

Title: Applications of Artificial Intelligence in Education

Blog:
Artificial Intelligence is transforming education by providing personalized learning,
automated grading, intelligent tutoring systems, and virtual classrooms. AI helps
teachers identify students' strengths and weaknesses while improving learning efficiency.

Now write a 200-word blog on 'Applications of Artificial Intelligence in Healthcare'.
```

```
"""

response = client.responses.create(
    model="gpt-5.4-mini",
    input=one_shot
)

print("\n===== ONE-SHOT OUTPUT =====")
print(response.output_text)


# ---------- Few-shot Prompt ----------
few_shot = """
Example 1

Title: Artificial Intelligence in Banking

Blog:
AI improves fraud detection, customer support through chatbots, risk analysis,
and personalized financial services.

Example 2

Title: Artificial Intelligence in Agriculture

Blog:
AI helps farmers monitor crops, predict weather conditions, detect plant diseases,
and improve crop yield using smart technologies.

Now write a 200-word blog on 'Applications of Artificial Intelligence in Healthcare'.
"""

response = client.responses.create(
    model="gpt-5.4-mini",
    input=few_shot
)

print("\n===== FEW-SHOT OUTPUT =====")
print(response.output_text)
```

Output:

===== ZERO-SHOT OUTPUT =====

Artificial Intelligence (AI) is transforming healthcare in powerful ways, making diagnosis, treatment, and patient care more efficient and accurate. One of the most important applications of AI is in medical imaging. AI systems can analyze X-rays, MRIs, and CT scans quickly, helping doctors detect diseases such as cancer, fractures, and lung conditions at an early stage.

AI is also widely used in predictive analytics. By studying patient records, lifestyle data, and genetic information, AI can help predict the risk of diseases like diabetes, heart disease, and stroke. This allows healthcare providers to take preventive action before conditions become serious.

Another major use of AI is in personalized treatment. AI can assist doctors in choosing the most effective medicines and therapies based on a patient's unique health profile. In addition, virtual health assistants and chatbots are improving patient support by answering questions, scheduling appointments, and providing basic medical guidance.

AI is also helping hospitals manage operations more smoothly by reducing paperwork, organizing patient data, and improving resource allocation. As technology continues to advance, AI will play an even bigger role in making healthcare faster, smarter, and more accessible for everyone.

===== ONE-SHOT OUTPUT =====

Applications of Artificial Intelligence in Healthcare

Artificial Intelligence is revolutionizing healthcare by improving diagnosis, treatment, and patient care. AI-powered tools can analyze medical images such as X-rays, CT scans, and MRIs with remarkable accuracy, helping doctors detect diseases like cancer, pneumonia, and heart conditions at an early stage. This leads to faster decisions and better outcomes for patients.

AI is also being used in personalized medicine, where treatment plans are designed according to a patient's medical history, genetics, and lifestyle. In hospitals, intelligent systems assist in managing patient records, scheduling appointments, and predicting disease outbreaks. Virtual health assistants and chatbots provide quick responses to common medical questions, making healthcare more accessible.

Another important application is in remote patient monitoring. Wearable devices connected to AI can track heart rate, blood pressure, and sugar levels in real time, alerting doctors when abnormalities occur. This is especially useful for elderly patients and those with chronic illnesses.

Overall, Artificial Intelligence is making healthcare more efficient, accurate, and patient-centered. While human expertise remains essential, AI supports medical professionals in delivering better care and saving lives.

===== FEW-SHOT OUTPUT =====

Applications of Artificial Intelligence in Healthcare

Artificial Intelligence (AI) is transforming the healthcare industry by making diagnosis, treatment, and patient care faster and more accurate. One of the most important applications of AI in healthcare is medical diagnosis. AI-powered systems can analyze X-rays, CT scans, and MRI images to detect diseases such as cancer, pneumonia, and heart problems at an early stage. This helps doctors make better decisions and start treatment sooner.

AI is also widely used in personalized medicine. By studying a patient's medical history, genetic information, and lifestyle, AI can help doctors suggest treatments that are more effective for individual patients. In addition, AI-powered chatbots and virtual assistants provide 24/7 support to patients by answering health-related questions, reminding them to take medicines, and helping them book appointments.

Hospitals also use AI to manage administrative tasks such as maintaining records, predicting patient admissions, and optimizing staff schedules. This reduces workload and improves efficiency. Furthermore, AI is playing a key role in drug discovery by analyzing large amounts of data to identify potential medicines more quickly.

In the future, AI will continue to improve healthcare by making it more accessible, accurate, and patient-friendly.

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Result

The Zero-shot, One-shot, and Few-shot prompting techniques were successfully implemented using the OpenAI API to generate a 200-word blog on the applications of Artificial Intelligence in Healthcare. The generated content was relevant, coherent, and demonstrated the effectiveness of different prompting strategies in improving AI-generated responses.